Customer Sentiment Analysis in Airline Reviews: A Comparative Analysis of Optimized Machine Learning, and NLP Techniques.

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***Abstract*—The rapid growth of online travel platforms and airline review websites has created a massive amount of unstructured customer feedback, making sentiment analysis increasingly important for improving passenger experience and service quality in the aviation industry. However, many existing studies focus on a single dataset or airline, sentiment categories such as positive, negative, and neutral opinions. In this paper, we propose an efficient framework for multi-score airline sentiment classification using three real-world datasets: the Skytrax User Review Dataset, the British Airline Reviews Dataset, and the Global Airlines Reviews Corpus. The proposed framework applies a complete Natural Language Processing (NLP) pipeline including text normalization,lowercase conversion, stopword removal with negation preservation, lemmatization, TF-IDF feature extraction, and ANOVA-based feature seelection. In addition, five machine learning algorithms are evaluated, namely Logistic Regression (LR), Decision Tree (DT), Random Forest (RF), Naïve Bayes (NB), and Stochastic Gradient Descent (SGD). Experimental results based on Accuracy, Precision, Recall, and F1-score demonstrate that Logistic Regression and Random Forest achieved the most consistent and reliable performance across the three datasets. The findings confirm that a well structured machine learning framework can effectively classify airline customer sentiment at scale while maintain strong generalization across review sources.**
Keywords: Sentiment Analysis, Airline Reviews, Natural Language Processing (NLP), TF-IDF, Logistic Regression, Decision Tree, Random Forest, Naïve Bayes, SGD, Tokenization, Lemmatization, Feature Engineering, ANOVA, Model Evaluation, Confusion Matrix, Multi-class Classification, Text Mining, Opinion Mining, Skytrax, British Airways, Global Airlines.

I. Introduction

Before trying a new restaurant, buying a product online, or booking a hotel, most people do one thing they check the reviews. This has become such a natural part of how we make decisions that we barely even notice we are doing it anymore. And on the other side of that equation, businesses are sitting on mountains of customer feedback that they genuinely do not know what to do with. A popular product on Amazon might have thirty thousand reviews. A hotel on TripAdvisor might have accumulated years worth of guest opinions.

This is where sentiment analysis comes in. The basic idea is straightforward instead of a human reading every review and deciding whether the customer was happy or not, you train a computer to do it. The computer reads the text, picks up on the language patterns, and assigns a label positive, negative, or neutral. Simple enough in theory. In practice, it turns out to be quite complicated, because people do not write reviews the way computers like to process them.

Customer feedback is messy in ways that are easy to underestimate. People abbreviate, they exaggerate, they use sarcasm, they contradict themselves within the same sentence. Someone writes "the delivery was surprisingly fast but the product itself was a complete letdown" and suddenly a simple positive or negative label does not really capture what that person experienced. Others throw in completely irrelevant information, use informal language that no grammar textbook would recognize, or switch between complaining and complimenting in the same paragraph. These are not edge cases this is just how people write when they are sharing a genuine opinion. And that gap between how humans naturally express themselves and how machine learning models need data to be structured is one of the central challenges this field has been working to close for years.

The research community has responded to this with a growing body of work in Natural Language Processing and Machine Learning, developing techniques to clean, structure, and extract meaning from raw text before any model ever sees it. Tokenization, stopword removal, lemmatization, TF-IDF vectorization these are the tools researchers have built specifically to bridge that gap. And they work, to varying degrees, depending on how carefully they are applied and how well they suit the domain at hand.

For this project, we focused on one domain in particular airline travel. It is a domain where customer feedback is both plentiful and genuinely important. Passengers who have just spent eight hours on a long haul flight tend to have strong opinions, and they share them. Platforms like Skytrax exist almost entirely for this purpose, accumulating hundreds of thousands of reviews covering everything from legroom and meal quality to how the cabin crew handled a difficult situation at thirty thousand feet. Airlines, for their part, could learn an enormous amount from this feedback but most of it sits unread because there is simply too much of it to process manually.

So we built a system to do it automatically. Starting from raw review text across three separate datasets the Skytrax User Reviews Dataset the British Airways Reviews Dataset, and the Global Airline Reviews corpus we designed an NLP pipeline to clean and prepare the text, applied TF-IDF vectorization to turn words into numbers, and trained five machine learning models to classify each review as positive, negative, or neutral. The five models we tested are Logistic Regression, Decision Tree, Random Forest, Naïve Bayes, and the SGD Classifier.

The main contributions of this work are as follows:
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- Developing a complete NLP preprocessing pipeline for customer review text, including negation aware stopword removal to preserve the meaning of phrases like "not good" or "never again."
- Applying TF-IDF vectorization with unigrams and bigrams to convert raw review text into structured numerical features.
- Training and comparing five machine learning classifiers across three independent real-world datasets.
- Evaluating all models using Accuracy, Precision, Recall, and weighted F1-Score to account for the class imbalance present across sentiment categories.
- Identifying which aspects of the flight experience such as cabin staff service and value for money most consistently drive positive or negative passenger sentiment.

The rest of this paper is organized as follows section 3 focuses on reviews related work and previous studies on sentiment analysis. Section 4 describes the datasets used, the preprocessing steps applied, and the full NLP pipeline. Section 5 presents the machine learning models and evaluation methods and discusses the experimental results and comparative analysis across all three datasets. Section 6 concludes the paper and outlines directions for future work.

II. Related Work

This study [1] proposes a Twitter-based sentiment analysis system called TweetAdvisor to measure customer satisfaction. Using the Support Vector Machine algorithm with unigram feature extraction, the system classifies tweets as positive or negative. The methodology includes data collection via Twitter, preprocessing, feature extraction, and SVM classification. The system also computes engagement metrics and supports comparative keyword analysis. Results on a 4,000-tweet dataset achieved 87% classification.

This reseacrh paper [2] applies NLP-based sentiment analysis to a Women's E-Commerce Clothing Reviews dataset to identify the correlation between customer reviews and product recommendations. Five machine learning algorithms were evaluated: Logistic Regression, SVM, Random Forest, XGBoost, and LightGBM. TF-IDF vectorization was used for text feature extraction. LightGBM achieved the best performance with 0.98 accuracy and 0.96 AUC, outperforming all other models. The findings provide insights into customer sentiment in the e-commerce apparel industry.

This study [3] proposes a customer sentiment analysis framework combining sentiment propagation and customer review analysis for automobile products. Semi-supervised learning applied to a word graph constructed via word embedding expands sentiment lexicons to capture domain-specific language. Reviews are clustered into major complaint topics, and three indices — controversy, complaint, and dissatisfaction — are designed to measure negative customer feedback. The method was tested on 311,550 reviews from ten online communities. Results demonstrate improved domain adaptability compared to standard sentiment dictionaries.

This research paper [4] presents a lexicon-based sentiment classification system for hotel customer reviews using TripAdvisor data. A domain-specific lexicon was constructed from a training corpus without external data sources, and sentiment scores were computed to classify reviews into five or three categories. Evaluation on 80,000+ reviews showed that the three-class model achieved approximately 90precision for positive and negative classifications. The study highlights that domain-specific lexicons outperform general-purpose ones. Future improvements include larger training sets and handling of negation and intensifiers.

This research paper [5] presents an aspect-based sentiment analysis approach integrated into a Customer Loyalty Improvement Recommender System (CLIRS) for heavy equipment repair services. The methodology combines opinion dictionaries (SentiWordNet, AFINN), syntactic dependency analysis, and aspect identification to extract structured sentiment from free-form customer feedback. Proposed improvements increased classification accuracy from 92% to 96% and coverage from 36% to 48%. The sentiment output is transformed into actionable knowledge using action rules to generate business recommendations. Remaining challenges include handling implicit opinions, pronouns, and idiomatic expressions.

This paper [6] presents a Hybrid Recommendation System that uses machine learning to analyze customer shopping experiences on online platforms. The system combines collaborative filtering and product similarity techniques to generate personalized recommendations. Key features such as price, quality, and customer reviews are used for classification. Performance evaluation was conducted using MAE, MSE, and MAPE metrics. Results showed an accuracy of nearly 98%, outperforming existing recommendation approaches.

This study [7] analyzed 14,880 TripAdvisor restaurant reviews in Pattaya from 2017–2022 using the VADER sentiment analysis model to examine customer opinions before and during the COVID-19 pandemic. The results showed a major decline in restaurant reviews during the pandemic because of lockdowns, travel restrictions, and restaurant closures. Most reviews were positive overall, but negative sentiments increased noticeably during COVID-19. The study identified the main customer complaints as poor service and staff behavior, followed by food quality and taste issues. Frequent positive words included "good," "great," and "nice," while negative reviews often contained "bad" and "poor." The VADER model achieved 84.5% accuracy, proving effective for analyzing online restaurant reviews. The findings help restaurant owners understand customer needs and improve food quality, service, pricing, and hygiene for future crisis management and post-pandemic recovery.

This research paper [8] presents a sentiment analysis framework for Customer Relationship Management (CRM) using Hierarchical Attention Networks (HAN) combined with an incremental learning mechanism. The system analyzes customer communications from multiple channels such as emails, instant messaging, websites, and social media to classify sentiments into positive, negative, and neutral categories. The methodology includes text preprocessing, word embedding generation, HAN-based deep learning classification, and retraining through operator feedback using sweep rehearsal to avoid catastrophic forgetting. The dataset contained over 30,000 annotated English and Italian customer requests collected from CRM services and public sentiment datasets. Experimental results showed that the proposed approach outperformed Naïve Bayes, lexicon-based methods, and commercial sentiment analysis services, achieving macro F1-scores of 0.89 for Italian and 0.79 for English. The study demonstrates that incremental learning can continuously improve CRM sentiment analysis performance without retraining the model from scratch.

This paper [9] proposes a hybrid sentiment analysis framework for gastronomy tourism focused on improving minority sentiment classification in customer reviews. The methodology combines data augmentation, feature engineering, machine learning classification, and business intelligence visualization techniques to address class imbalance problems commonly found in tourism datasets. Text preprocessing methods including tokenization, stop-word removal, stemming, and TF-IDF feature extraction were applied before training classification models. Several machine learning algorithms, including Support Vector Machine (SVM), Random Forest, and ensemble methods, were evaluated. The proposed framework significantly improved the detection of minority sentiment classes and enhanced overall classification performance compared to traditional approaches. Business intelligence dashboards and visualization tools were also integrated to support tourism industry decision-making and customer satisfaction analysis. The study demonstrates that combining data augmentation and advanced feature engineering can effectively improve sentiment analysis accuracy in gastronomy tourism applications.

This study [10] investigates customer satisfaction with Traveloka, an Indonesian mobile travel application, through sentiment analysis of Twitter data. A dataset of 1,200 tweets was collected via the Twitter API, filtering Indonesian posts containing Traveloka-related keywords. Text data was transformed into numerical form using TF-IDF vectorization, then classified using three machine learning models: Support Vector Machine (SVM), Logistic Regression, and Naïve Bayes. Results indicate that 610 tweets were positive and 590 negative, with positive sentiment largely linked to promotions and discounts. SVM achieved the highest classification accuracy at 84.58%, , outperforming Logistic Regression (82.50%) and Naïve Bayes (82.91%).

This study [11] proposes a sentiment analysis framework for airline customer reviews using multiple machine learning models and Google's BERT language representation model. The dataset was sourced from Kaggle and comprised airline passenger reviews categorized as positive, negative, or neutral. Traditional ML algorithms including Logistic Regression, SVM, Decision Tree, Random Forest, KNN, and Adaboost were tested and compared against BERT. Word clouds and bar graphs were employed to identify the primary causes of negative feedback, such as customer service issues and flight cancellations. BERT achieved the highest accuracy of 83%, outperforming all ML models including the best-performing Random Forest (77%) across all metrics — accuracy, precision, recall, and F1-score.

This study [12] proposes an automated method for clustering users of a digital tourism platform based on sentiment analysis of their reviews. The methodology employs a multilingual lexicon-based sentiment classification algorithm combined with K-Means clustering to group users by sentiment polarity. Pre-processing techniques such as tokenization, stop-word removal, and language detection are applied before sentiment scoring. The system was tested on the Cuscarias platform across four languages, achieving accuracies of 91% (Portuguese), 89% (English), 75% (Spanish), and 71% (French). Results enabled effective user segmentation by age, gender, nationality, and sentiment, supporting data-driven tourism service improvements.

This paper [13] presents a systematic literature review on sentiment analysis in social media and its real-world applications. The study reviewed research published between

2014 and 2019 from major academic databases including ACM, IEEE Xplore, ScienceDirect, Emerald Insight, and Scopus. Out of 77 collected papers, 24 studies were selected after detailed screening and analysis. The review examined commonly used sentiment analysis methods, social media platforms, and application domains. Results showed that opinion-lexicon approaches were the most frequently applied techniques, while Twitter was the dominant data source for sentiment analysis research. Applications of sentiment analysis were identified in multiple fields including healthcare, politics, business, and world events. The paper concludes that social media sentiment analysis is an important tool for extracting valuable insights from large-scale user-generated content and highlights the growing role of big data analytics in opinion mining research.

This research paper [13] examines sentiment analysis in social media by analyzing 24 studies published between 2014 and 2019, sourced from five academic databases. The review addresses three questions covering methods used, social media platforms, and application contexts. Findings show that Twitter is the dominant data source, used in 85% of reviewed studies, while the opinion-lexicon method is most commonly applied. Sentiment analysis applications span world events, healthcare, politics, and business, with combining lexicon-based and machine learning methods recommended for improved accuracy.

This study [14] published in Mathematics (2023/2024) explored the use of BERT and other transformer-based models for airline sentiment classification using Twitter data. The research highlighted how AI and NLP techniques can help the aviation industry better understand customer opinions shared on social media, ultimately supporting improved decision-making and operational efficiency. During preprocessing, the researchers used LabelEncoder to prepare the dataset for classification tasks. The findings showed that BERT-based models achieved stronger performance than traditional machine learning approaches because they are better at understanding context and language meaning. However, these models also require greater computational power and training resources. For this reason, traditional methods such as Logistic Regression and SGD are still considered effective and practical options in environments with limited resources, which aligns well with the machine learning approach adopted in our project.

This paper [15]published in December 2024, compares machine learning and deep learning models for sentiment analysis using Amazon customer reviews. The study evaluated deep learning approaches such as CNN and RNN alongside traditional models including Logistic Regression, Random Forest, and Naive Bayes. Using star ratings as sentiment labels, the research highlighted the issue of class imbalance due to the dominance of positive reviews, similar to the underrepresented Neutral class in our project. Results showed that deep learning models achieved higher accuracy on large datasets, while Logistic Regression remained highly efficient with lower computational requirements. The findings support our use of star ratings as sentiment indicators and validate the selection of Logistic Regression, Random Forest, and Naive Bayes for model comparison.

This study [16] published in February 2024 by researchers from Purdue University, explores the use of machine learning for airline passenger sentiment analysis using Twitter data. The research combines NLP, sentiment analysis, lexical analysis, and time-series techniques to analyze passenger opinions and detect sentiment trends over time. Text vectors were processed using pre-trained machine learning classifiers, while keyword frequency analysis provided additional contextual understanding of customer feedback. The study also applied Bollinger Bands to identify sudden sentiment changes, showing that traditional ML models such as Naive Bayes, Logistic Regression, Random Forest, and SVM effectively detect shifts in passenger sentiment. These findings directly support our project's use of multiple machine learning algorithms for practical airline sentiment monitoring.

This paper [17] applies NLP, text analysis, biometrics, and computational linguistics techniques to analyze sentiments and emotions in the Skytrax Airline Customers' Feedback (SACF) dataset. The study evaluated several deep learning models, finding that LSTM achieved the highest accuracy at 91%, followed by BERT transfer learning at 90% and Conv1D at 87%. The results confirmed that deep learning methods are highly effective for sentiment analysis, while also highlighting difficulties when working with smaller domain-specific datasets. This research is highly relevant to our project because it uses the same Skytrax dataset and a similar NLP preprocessing pipeline, providing a direct benchmark for comparison. Our traditional machine learning approach using TF-IDF delivers competitive performance while requiring significantly lower computational resources than the LSTM and BERT models used in the study.

This paper [18] published in the Natural Language Processing Journal in March 2024, surveys sentiment analysis techniques ranging from traditional machine learning models to deep learning and transformer-based approaches. The study examines common NLP preprocessing methods such as tokenization, stopword removal, stemming, and lemmatization, which align with our project's preprocessing pipeline. It evaluates algorithms including Naive Bayes, SVM, Logistic Regression, Random Forest, CNN, LSTM, GRU, and transformers using standard evaluation metrics like accuracy, precision, recall, and F1-score. The review concludes that although transformer models provide the highest accuracy, traditional machine learning methods combined with TF-IDF remain highly effective with much lower computational cost. It also identifies neutral sentiment classification as a major ongoing challenge, consistent with the results observed in our project datasets.

This paper [19] focuses on analyzing airline-related Twitter posts using sentiment analysis techniques. The study applies several NLP preprocessing methods such as text cleaning and stop-word removal to prepare the textual data. TF-IDF vectorization was used to convert text into numerical features suitable for machine learning models. Multiple algorithms including Logistic Regression, SVM, Random Forest, and Naive Bayes were implemented and compared. The models were evaluated using different classification performance metrics. Experimental results showed that traditional machine learning models can achieve high sentiment classification accuracy when combined with effective preprocessing methods. The paper concluded that sentiment analysis can provide valuable insights into customer opinions toward airline services.

This research paper [20] investigates sentiment analysis techniques for airline customer reviews. The study highlights the importance of customer feedback in evaluating airline service quality and customer satisfaction. NLP preprocessing techniques were applied to clean and analyze textual review data collected from online platforms. Several machine learning methods were used to classify customer sentiments and identify important review features. The research examined different airline service aspects such as comfort, staff behavior, and service quality. Experimental findings showed that sentiment analysis models can effectively classify airline reviews into sentiment categories. The paper concluded that automated sentiment analysis can help airlines improve their services and better understand passenger opinions.

In this study [21] a Machine Learning Based Approach with TF-IDF Vectorization", presents a machine learning framework for analyzing customer reviews. The study emphasizes the role of sentiment analysis in understanding customer opinions and supporting business decisions. Several NLP preprocessing operations such as tokenization, normalization, and stop-word removal were applied to improve data quality. TF-IDF vectorization was used to transform textual reviews into numerical feature representations. Different machine learning classification algorithms were implemented and evaluated for sentiment prediction. The experimental results demonstrated that combining TF-IDF with machine learning models achieves strong classification performance. The paper concluded that sentiment analysis systems can efficiently process and analyze large volumes of feedback.

This research paper [22] compares traditional machine learning and deep learning techniques for sentiment analysis tasks. The study focuses on classifying text into positive, negative, and neutral sentiments using NLP methods. Several preprocessing techniques were applied to clean and prepare the textual datasets before model training. Traditional machine learning algorithms such as Logistic Regression and SVM were compared with deep learning models. The researchers also analyzed the effect of feature extraction methods including TF-IDF and word embeddings. Experimental results showed that deep learning performs well on large datasets, while classical machine learning remains efficient and effective for many tasks. The paper concluded that model selection depends on dataset size, computational resources, and application requirements.

This paper [23] provides a comprehensive review of the development of sentiment analysis research over time. The study discusses the growing importance of sentiment analysis in understanding opinions expressed in online reviews, social media, and customer feedback platforms. It examines the major research topics, methodologies, and techniques commonly used in sentiment analysis applications. The paper also reviews the evolution of Natural Language Processing and machine learning approaches used for opinion mining tasks. Different sentiment classification methods, datasets, and evaluation techniques were analyzed and compared. The study highlights the increasing use of deep learning and advanced NLP models in recent sentiment analysis research. The paper concluded that sentiment analysis continues to evolve rapidly and remains an important research area with applications in many real-world domains.

This research paper [24] focuses on analyzing online airline customer reviews using supervised machine learning techniques for sentiment analysis and opinion mining. The study aims to classify customer opinions into positive, negative, and neutral sentiments to better understand passenger satisfaction and service quality. Several preprocessing techniques such as text cleaning, tokenization, and stopword removal were applied to improve the quality of textual data before classification. The research used feature extraction methods to transform customer reviews into numerical representations suitable for machine learning models. Multiple supervised machine learning algorithms were implemented and compared to evaluate their effectiveness in sentiment classification tasks. The experimental results showed that some models achieved high accuracy in predicting customer sentiment from airline reviews. The paper concludes that machine learning and NLP techniques can significantly help airlines analyze customer feedback automatically and improve decision-making and service performance.

This paper [25] presents a systematic review of sentiment analysis and topic modeling techniques used in the airline industry to analyze customer opinions and experiences. The study examines how Natural Language Processing (NLP) and Machine Learning methods help airlines understand passenger satisfaction through online reviews and social media data. It discusses various preprocessing techniques such as text cleaning, tokenization, stopword removal, and feature extraction methods commonly used in sentiment analysis tasks. The paper also reviews different machine learning and deep learning models applied for sentiment classification and opinion mining in airline-related datasets. In addition, topic modeling techniques are explored to identify the most discussed issues among airline customers, such as delays, food quality, and customer service. The review highlights the strengths and limitations of existing approaches and compares their performance across different studies. The paper concludes that sentiment analysis and topic modeling can significantly support airlines in improving customer satisfaction, service quality, and decision making processes.

III. Proposed Methodology

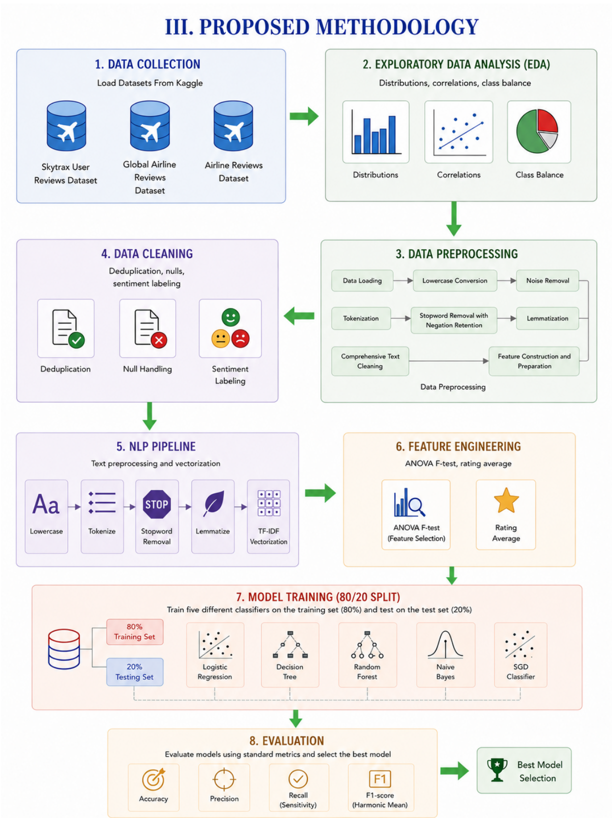

Fig. 1: Airline Reviews Decompose

A. Datasets Descriptions

This Study uses three airline review datasets from Kaggle for sentiment analysis and text classification in natural language processing (NLP). Each dataset contains passenger reviews with ratings and text feedback, providing a supervised learning approach to analyzing customer sentiment towards airlines.

The first dataset is the Airline Reviews Dataset, which focuses specifically on British Airways customer feedback. It includes review text, ratings for various services,traveler type,seat class,and route information.

The second dataset is the Skytrax User Reviews Dataset, which contains user reviews scraped from Skytrax covering airlines, airports,lounges,and seats. it offers a wide range of passenger feedback across different service areas.

The third dataset is the Global Airline Reviews dataset, which includes passenger reviews for airlines worldwide. it provides text reviews along with detailed ratings for seat comfort,staff service,food entertainment,and value for money.

All three datasets contain a mix of positive and negative sentiments,making them suitable for training and evaluating sentiment analysis models. They also support text preprocessing,feature extraction,and classification tasks.

TABLE I: Features of Airline Review Datasets

Feature	Type	Description
Review Text	Text	The full text of the passenger's review
Overall Rating	Numerical	Overall score given by the passenger (1-5 or 1-10)
Recommended	Categorical	Whether the passenger recommends the airline (Yes/No)
Type of Traveller	Categorical	Business, Leisure, Couple, Family, etc.
Seat Type	Categorical	Economy, Business, First Class
Route	Text	Flight origin and destination
Seat Comfort	Numerical	Rating for seat comfort
Cabin Staff Service	Numerical	Rating for cabin crew service
Value for Money	Numerical	Rating for flight value

B. Data Preprocessing

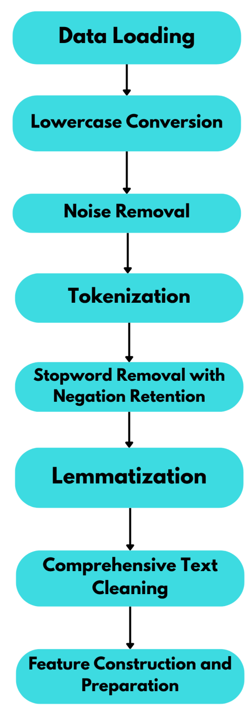

Fig. 2: Preprocessing Steps

Data preprocessing plays a critical role in our Customer sentiment analysis project. The reason behind this is that the data we collected is unstructured and contains errors. The data is collected from sources such as customer reviews and airline opinions. Data preprocessing ensures that the data is available for use in Machine Learning algorithms. Preprocessing includes removing punctuation symbols and special characters, stopwords, and duplicate entries. After that , we conducted tokenizarion and lemmatization.

1) **Data Loading:** The dataset was loaded into Google colab KaggleHub connector. The Skytrax User Reviews dataset was imported in csv format and configured with selected columns such as review, sentiment, seatComfort, cabin staff service. for safety, a working copy of each dataset (e.g., df_processed,df3_clean,df_clean_mobile) was created to perform all processing steps consistently.

2) **Lowercase Conversion:** All review was converted to lowercase to ensure consistency and prevent discrepancies during token matching. This ensures that words like "Good" and "good" are treated identical , improving the quality of text comparison.

3) **Noise Removal:** Noise Removal: Reviews collected from online platforms rarely come clean they contain URLs, email addresses, numbers, punctuation, and special characters that have nothing to do with how a passenger actually felt. Regular expressions were applied to remove all of these, leaving behind only the words that genuinely carry sentiment meaning.

4) **Tokenization:** Each review sentence was broken down into individual words using the word_tokenize() function from NLTK. Tokenization allows the text to be processed at the word level in later steps like stopword removal and lemmatization

5) **Lemmatization:**
Lemmatization was applied using WordNet Lemmatizer to reduce words to their base form. This means words like "running," "ran," and "runs" all become "run." this step helps reduce word variation and improves text representation in the model.

6) **Stopword Removal with Negation Retention:** we get rid of words like "is" "the" "and" because they do not really add much meaning we

7) **Comprehensive Text Cleaning:** After all transformations, the cleaned tokens were joined together into smooth text sentences and saved as a new tect column (Review_cleaned, clean_review ,or review_cleaned). The outputs were verified by comparing original and processed text to ensure sentimentrelevant information was preserved.

8) **Feature Preparation:** Additional features were computed to enrich the dataset including review length and word count for basic text statistics. For the airline dataset, aggregated rating futures like overall rating mean and rating standard deviation were calculated. Finally, the textual sentiment column was encoded into numeric form using LabelEncoder, and reviews were transformed into vectorized numerical features using TF-IDF Vectorizer with Unigrams and Bigrams. The resulting matrix represented the structured input prepared for machine_learning and sentiment_classification stages.

C. Dataset visualization

Data visualization greatly simplifies working with data, especially in domains such as sentiment and customer-feedback analysis. By translating complex quantitative and textual evaluations into clear, visual figures, it becomes possible to detect patterns of satisfaction, service quality, and emotional tone. Without such visualization, it would be difficult to recognize distinctions between satisfied and unsatisfied passengers in a large corpus of reviews. Through the following visual summaries, the dataset's structure, quality, distributions, and sentiment balance were examined before modeling.

1) **Sentiment Distribution:** The target variable "recommended" served as a proxy for customer sentiment. Frequency plots and bar charts showed ratio of positive (yes) and negative (no) recommendations. Across datasets,the proportion of positive recommendations was dominant, with approximately 70-80% of entries marked as "yes," indicating general satisfaction among reviewers.

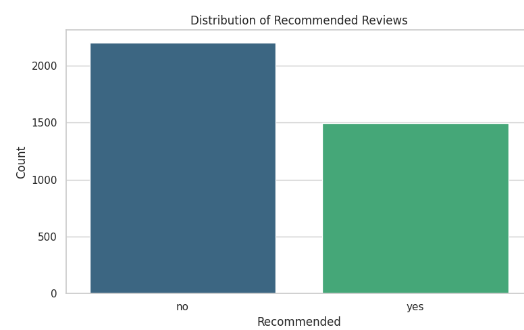

Fig. 3: Sentiment Class Distribution of British Airways

2) **Average Rating Distribution:** An overall rating proxy was derived by averaging seven numeric sub_ratings, seat comfort, cabin staff service, food and beverages, inflight entertainment, ground service, wifi and connectivity and value for money.

3) **Missing Values Overview:** A missing value bar chart summarized data completeness across all columns. Most datasets exhibited minimal null presence (<5%) with slightly more missing entries in text heavy or optional fields like review body or wifi and connectivity. The visualization guided selective imputation and informed which columns were excluded from modeling.

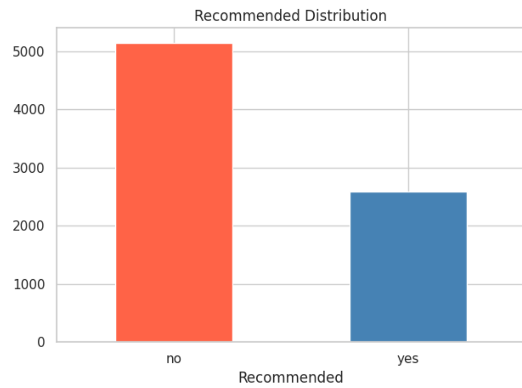

Fig. 4: Sentiment Class Distribution of Global Airways

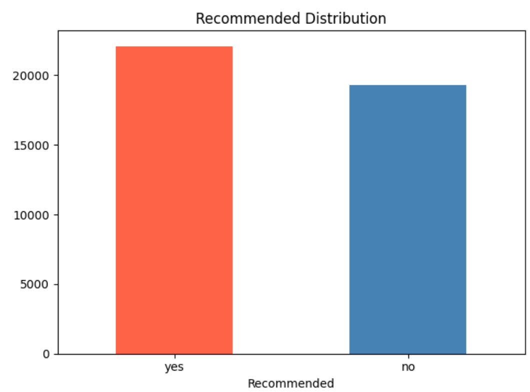

Fig. 5: Sentiment Class Distribution of Skytrax Airways

4) **Numeric Feature Distributions:** Boxplots grouped by recommended status were generated for all numeric rating features. This helped identify variation and potential outliers. Positively recommended reviews generally corresponded to higher median values across all attributes, especially for cabin staff service and seat comfort, pointing to their strong correlation with customer satisfaction.

5) **Categorical Feature Relationships:** Bar charts comparing type of traveler, seat type and verified review indicated differing likelihoods of recommendations. for example business class and leisure travelers presented a higher proportion of positive recommendations. verified reviews also leaned more positive, suggesting higher authenticity among satisfied travelers.

6) **Correlation Matrix:** correlation heatmap visualized linear relationships among numeric features and recommended. Key findings included strong positive interfeature correlations within comfort and service-related variables, and a moderate positive correlation (approx 0.55–0.65) between value for money and recommended. This indicates that customers equate perceived value directly with overall satisfaction.

7) **Review Word Count Distribution:** to examine the impact of preprocessing on the textual corpus, the word count distribution for each review was compared before and after cleaning. Words counts quantify the volume of linguistic information in a review and help assess data compactness after transformation. The raw data showed highly variable review length_ranging from just a few words to well over 100 words-due to punctuation, filler expressions, unnecessary tokens. After cleaning, the distribution became more consistent, with most reviews falling between 20 and 30 words, representing nearly a 50% reduction in length This shows that preprocessing successfully removed redundant information while keeping the main meaning of the reviews intact. As a result the cleaned text became more suitable for TF-IDF feature extraction and sentiment classification.

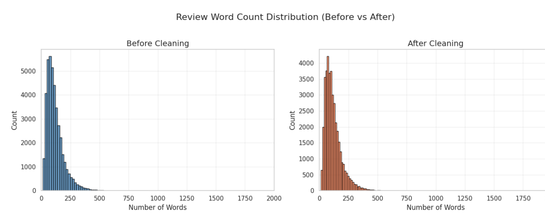

Fig. 6: Sentiment Class Distribution of Airlines

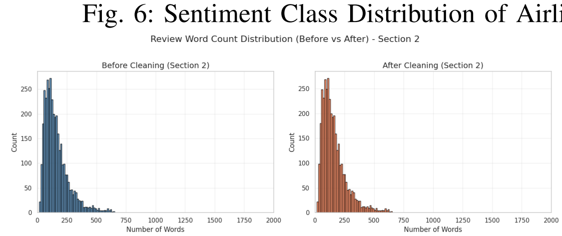

Fig. 7: Sentiment Class Distribution of British Airways

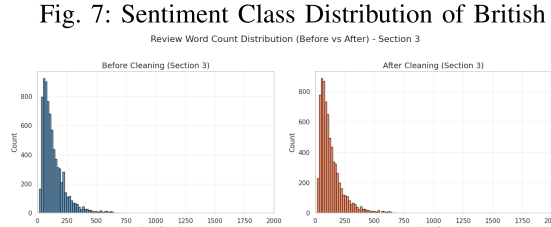

Fig. 8: Sentiment Class Distribution of Global Airlines

8) **Word Clouds:** Separate word clouds were created for positive and negative reviews based on sentiment derived from recommended. Positive reviews featured words such as "excellent," "friendly," "comfortable," "great service" whereas negative ones highlighted "delay," "rude," "poor,". The contrasting visuals immediately reflected the tone and key factors behind user perceptions.

Fig. 9: Word Cloud of Airlines

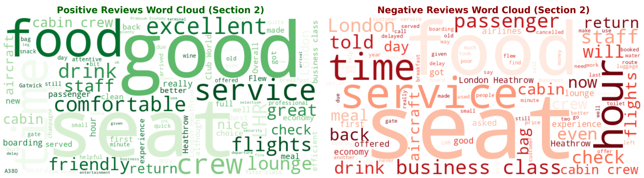

Fig. 10: Word Cloud of British Airways

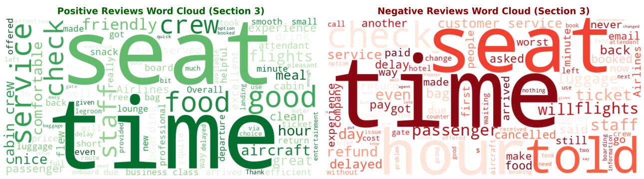

Fig. 11: Word Cloud of Global Airlines

9) **Bag-of-Words:** TF-IDF vectorization was applied using unigrams and bigrams to capture contextual meaning. The most frequent words across all reviews included service-related terms such as "staff," "seat," and "flight," while chi-square-based importance visualization underscored that words like "excellent" appeared predominantly in positive reviews and "delay" or "worst" in negative ones. These bar plots facilitated linguistic interpretation of sentiment polarity. Beyond individual word frequency, the bigram analysis revealed meaningful two-word combinations that single tokens alone could not capture — phrases like "great service," "friendly staff," and "comfortable seat" consistently dominated positive reviews, while "flight delayed," "rude staff," and "poor service" were strongly associated with negative sentiment. This confirmed that analyzing word pairs alongside individual words provides a richer and more nuanced understanding of how passengers express their opinions, making the combined unigram and bigram approach more effective than using either alone.

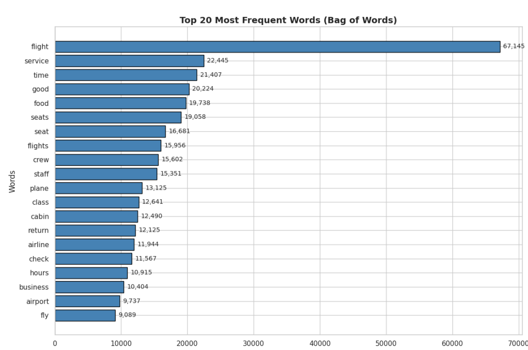

Fig. 12: Top 20 Words of The Airlines Dataset

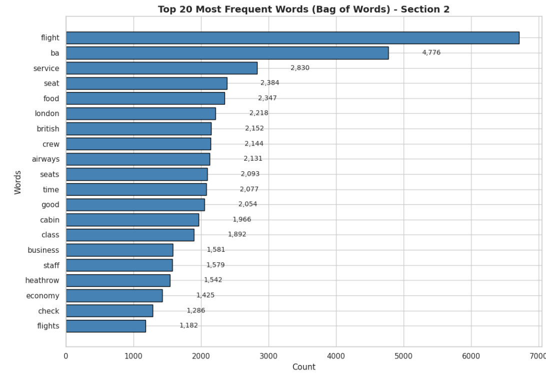

Fig. 13: Top 20 Words of The British Airways Dataset

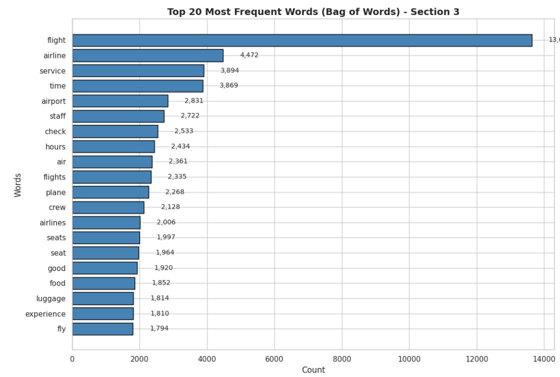

Fig. 14: Top 20 Words of The Global Airlines Dataset

10) **N-gram:** To capture more meaningful context, a bi-gram analysis was performed. Unlike single-word analysis, bi-grams focus on pairs of words that often appear together, providing clearer insights into customer opinions. Positive bi-grams such as "great service", "friendly staff", "comfortable seat", and "excellent flight" appeared most frequently. On the negative side, common phrases included "poor services", "rude staff", "delayed flight", and "lost luggage".

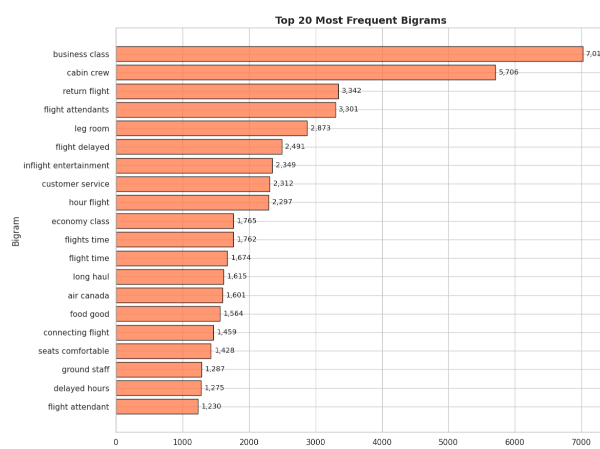

Fig. 15: Top 20 Bigrams of The Airlines Dataset

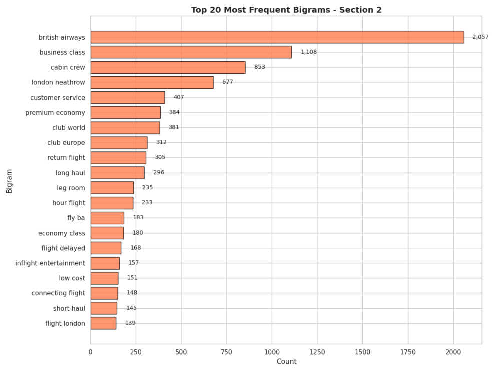

Fig. 16: Top 20 Bigrams of The British Airways Dataset

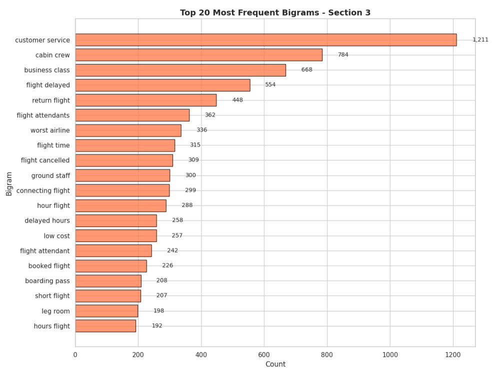

Fig. 17: Top 20 Bigrams of The Global Airlines Dataset

11) **Sentiment Lexicon Overlap:** Sentiment Lexicon Overlap was conducted to check whether the review text matched the actual recommended labels in the dataset. This was done by counting positive and negative keywords using a predefined sentiment dictionary.

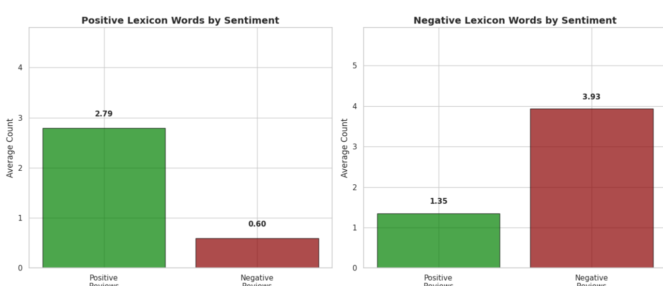

Fig. 18: Lexicon of The Airlines Dataset

Fig. 19: Lexicon of The British Airways Dataset

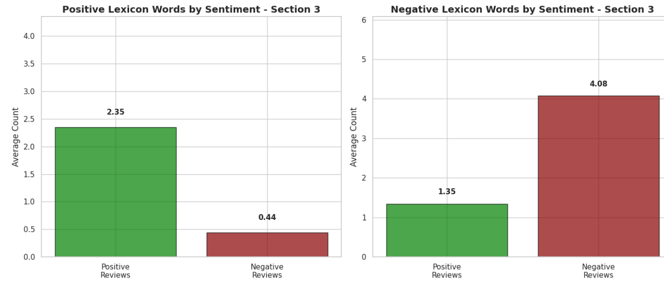

Fig. 20: Lexicon of The Global Airlines Dataset

D. *Used Algorithms*

As our goal was to test multiple machine learning models to select the most accurate and high-performance models for the sentiment analysis of Airline passenger reviews,we used traditional machine learning models **(Random Forest, SGD Classifier, Naive Bayes,Logistic Regression, Decision Tree)** However, all these models were evaluated based on accuracy, speed, and efficiency.

a) **Decision Tree:**
Decision Tree is one of the machine learning models we used in our project for airline passenger review sentiment analysis. It is a type of supervised learning that tries to find the best splits in the data based on feature values to separate reviews into sentiment classes.The primary goal of the Decision Tree is to select splitting conditions that maximize the separation between classes at each node.In our study, we utilized the Decision Tree to determine whether a passenger review was positive, Neutral ,or Negative. We compared it to other models, evaluating its accuracy, precision, recall and F1-score across all three datasets. The Decision Tree gave interpretable results and provided a clear baseline for comparison with more complex models.

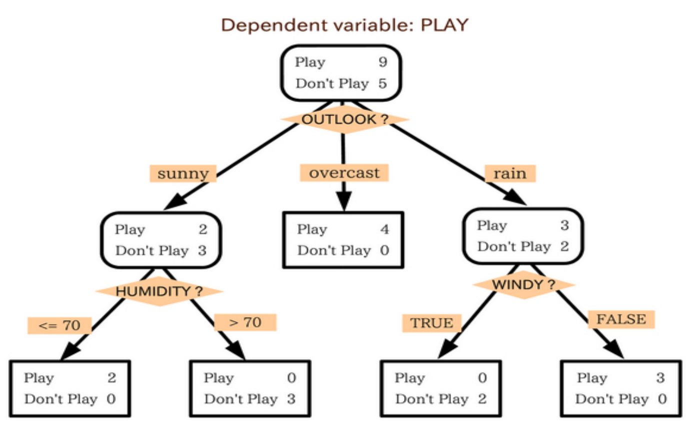

Fig. 21: Desicion Tree

b) **Logistic Regression:**
The Logistic Regression model is one of the most fundamental and widely used techniques for classification problems, and it is very effective in sentiment analysis of airline passenger reviews,It estimates the likelihood of a review expressing a positive ,neutral,or negative sentiment by applying a logistics function to a linear combination of the input features, which in this case are the TF-IDF weighted words extracted from the review text. Words such as "excellent","comfortable",and "friendly" are highly indicative of positive sentiment whereas "terrible","delayed",and "rude" are indicative of negative sentiment. Logistic Regression is widely used as it is simple,fast,and interpretable,and it performed well across all three datasets making it a strong and reliable baseline model.

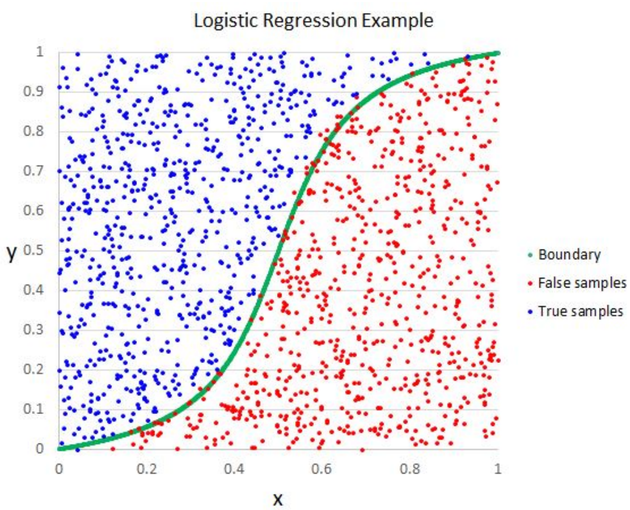

Fig. 22: Logistic Regression

c) **SGD Classifier:**
The SGD Classifier is a lightweight and robust linear model, that is highly effective for high-dimensional text classification tasks such as airline passenger review sentiment analysis.In our project, we used it to classify reviews as positive,neutral,or negative based on TF-IDF features extracted from the review text, It is very fast when working with large datasets since weights are updated incrementally after each training instance using stochastic gradient descent.Despite its simplicity,the SGD Classifier performed well and provided a reasonable and competitive result when compared to the other models across all three datasets.

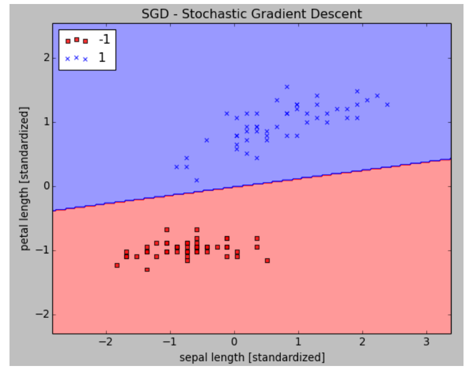

Fig. 23: SGD Classifier

d) **Random Forest :**
Random Forest works by building a large number of decision trees rather than relying on a single one, which is what separates it from a standard Decision Tree classifier. Each tree is trained on a randomly selected portion of the data and a random subset of features — a technique known as bagging — and the final prediction is determined by combining the output of all trees together. In this project, TF-IDF features extracted from passenger reviews were used to train the model to classify sentiment as positive, neutral, or negative across the Skytrax, British Airways, and Global Airlines datasets. The reason this model suits our task well is that passenger reviews frequently contain mixed opinions and inconsistent language that create non-linear patterns in the data, and a single decision tree simply cannot capture that kind of complexity reliably. By aggregating predictions across many trees, Random Forest naturally reduces variance and produces more stable results. It requires more computational time compared to simpler models, but the improvement in handling the linguistic diversity present across three different datasets made it a justified and dependable choice for this comparison.

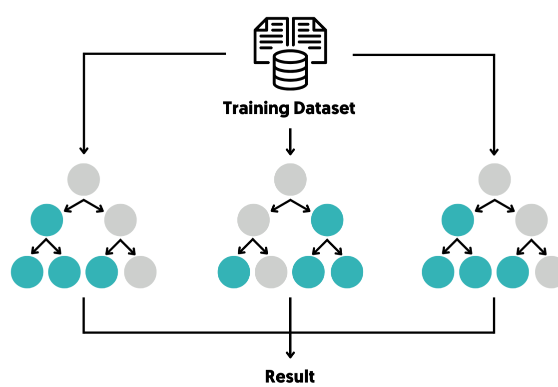

Fig. 24: Logistic Regression

e) **Naive Bayes:**
Naive Bayes is a probabilistic model that assumes every feature is conditionally independent of the others. It is particularly well suited for text classification tasks as it works effectively with TF-IDF based features derived from review text. In our sentiment analysis project, we used Multinomial Naive Bayes which is very effective for count-based and frequency-based text features. We applied it across all three airline review datasets to classify reviews into Positive, Neutral, or Negative sentiment classes. Naive Bayes is an efficient and fast classifier that provided a strong and competitive baseline despite its simplicity because it efficiently handles high-dimensional sparse text data while maintaining low computational cost.

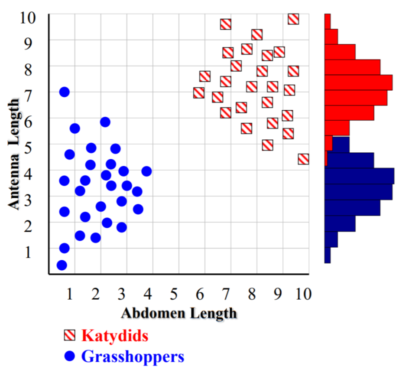

Fig. 25: Naive Bayes

E. *Performance Metrics*

A range of evaluation metrics are used to measure the performance of predictive models. Accuracy provides an overall measure of model performance by calculating the proportion of correct predictions, including both true positives and true negatives. Precision quantifies the accuracy of positive predictions by measuring the proportion of true positive predictions relative to all predicted positive cases. Recall, on the other hand,evaluates the model's ability to correctly identify actual positive cases by comparing true positives with the total number of real positive instances. Together, these metrics provide different perspectives on model performance and help researchers understand the effectiveness and limitations of predictive models.

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}} \tag{1}$$

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \tag{2}$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{3}$$

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \tag{4}$$

$$\text{FPR} = \frac{\text{FP}}{\text{FP} + \text{TN}}, \quad \text{TPR} = \frac{\text{TP}}{\text{TP} + \text{FN}} \tag{5}$$

$$\text{Confusion Matrix} = \begin{pmatrix} \text{TP} & \text{FP} \\ \text{FN} & \text{TN} \end{pmatrix} \tag{6}$$

$$\text{Importance}_i = \text{Performance}_{\text{original}} - \text{Performance}_{\text{shuffled}(i)} \tag{7}$$

$$\text{Importance}_i = \|\vec{w}_i\| = \sqrt{\sum_{j=1}^{d} w_{ij}^2} \tag{8}$$

$$\text{Importance}_i = \sum_{s \in \text{Splits}(i)} \text{Gain}(s) \tag{9}$$

IV. Results and Analysis

In this section, the results achieved after the application of five machine learning algorithms on three airline customer reviews' datasets are shown. To make sure that a level playing field is maintained when comparing the performance of these machine learning algorithms against each other, a common pipeline was adopted for all datasets, which included data cleaning, natural language processing, TF-IDF vectorization, and training.The efficiency of these algorithms was calculated by four different metrics, namely Accuracy, Precision, Recall, and Weighted F1-Score.

A. *Dataset 1: Skytrax Airline Reviews*

The first dataset consists of 41,396 reviews collected from the Skytrax platform and covers reviews for 362 airlines. After preprocessing and removing duplicate entries, the dataset was divided into 31,756 training samples and 7,940 testing samples. Sentiment labels were generated based on the average of five service-related rating categories. The final distribution of sentiments was approximately 59 Negative, 23 Positive, and 18 Neutral, indicating a moderate class imbalance. To account for this imbalance, the weighted F1-Score was included as an important evaluation metric. Table II shows the performance results .

TABLE II: Model Performance on Dataset 1 — Skytrax Airline Reviews

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.6942	0.6762	0.6942	0.6804
SGD Classifier	0.6700	0.6790	0.6700	0.6583
Naive Bayes	0.6487	0.6206	0.6487	0.6154
Random Forest	0.6465	0.6239	0.6465	0.5607
Decision Tree	0.6000	0.5694	0.6000	0.5740

12) **Logistic Regression**

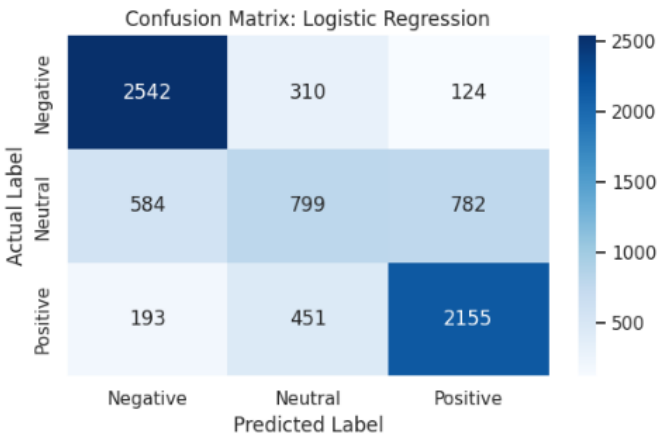

Fig. 26: Logistic Regression — Airline Reviews Dataset

13) **Decision Tree**

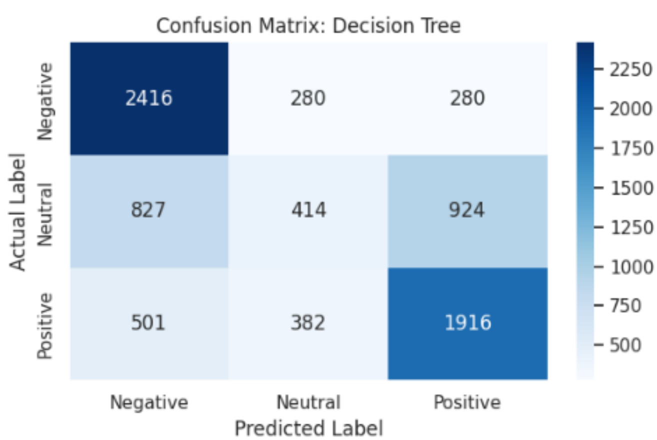

Fig. 27: Decision Tree — Airline Reviews Dataset

14) **Random Forest**

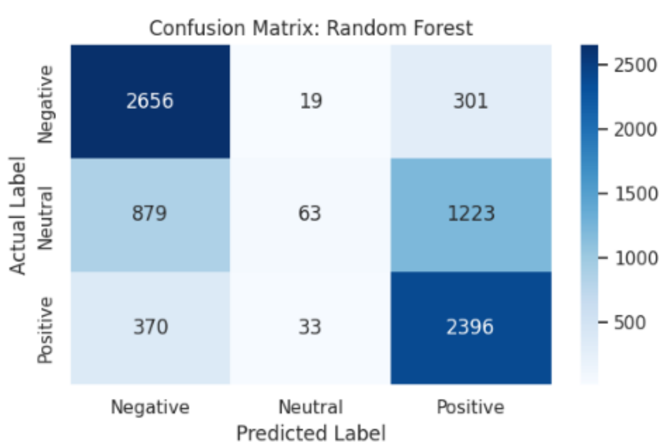

Fig. 28: Random Forest — Airline Reviews Dataset

15) **Naive Bayes**

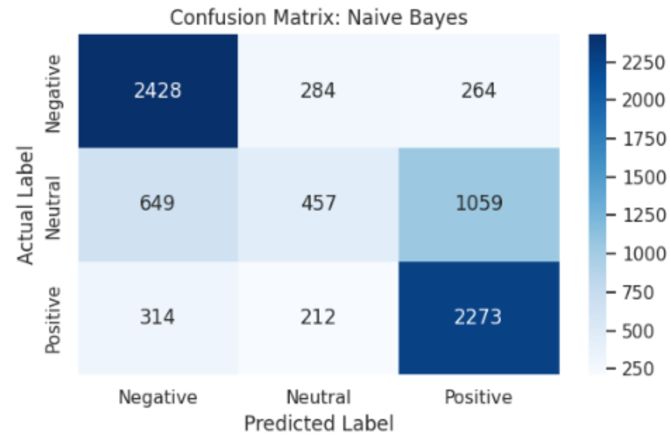

Fig. 29: Naive Bayes — Airline Reviews Dataset

16) **SGD Classifier**

Fig. 30: SGD Classifier — Airline Reviews Dataset

B. Dataset 2: British Airways Reviews

The second dataset consists of British Airways customer reviews sourced from Kaggle. Sentiments for these reviews were assigned based on four dimensions available for ratings, which are Seat Comfort, Cabin Staff Service, Food&Beverages, and Value For Money.

Both NLP preprocessing steps and TF-IDF model were used separately for this data set without any leakage from Dataset 1.

Table III reports the performance of all five models on Dataset 2.

TABLE III: Model Performance on Dataset 2 — British Airways Reviews

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	0.6509	0.6429	0.6509	0.6437
SGD Classifier	0.6225	0.6172	0.6225	0.6192
Naive Bayes	0.6279	0.6022	0.6279	0.5963
Random Forest	0.6238	0.6054	0.6238	0.5852
Decision Tree	0.5467	0.5322	0.5467	0.5247

For Dataset 2, results are somehow lower in comparision to dataset 1 across all models because the number of sentiment label dimensions for both negative and positive ratings was lower(four against 5) while the topic scope became narrower.

Once more, the best model in terms of F1-Score performance turned out to be logistic regression, with a result of 0.6437. The order of model results remained practically the same as in dataset 1.

The confusion matrices for all five models on Dataset 2 are presented in Figures 31 through 35.

Logistic Regression

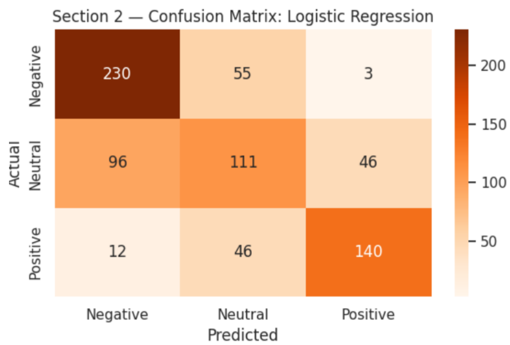

Fig. 31: Logistic Regression — British Airways Dataset

Decision Tree

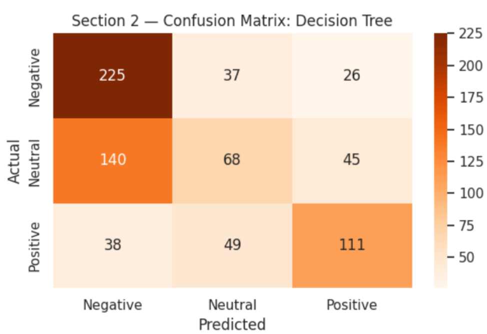

Fig. 32: Decision Tree — British Airways Dataset

Random Forest

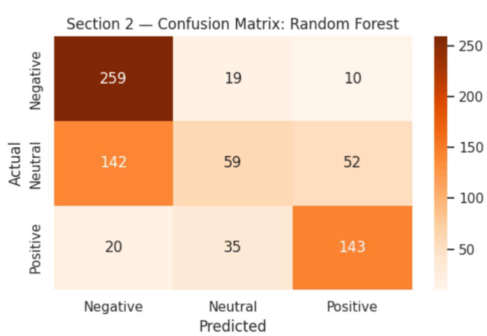

Fig. 33: Random Forest — British Airways Dataset

Naive Bayes

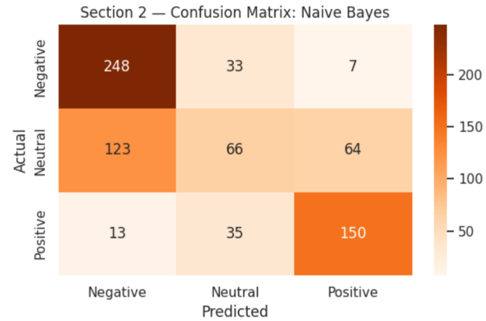

Fig. 34: Naive Bayes — British Airways Dataset

SGD Classifier

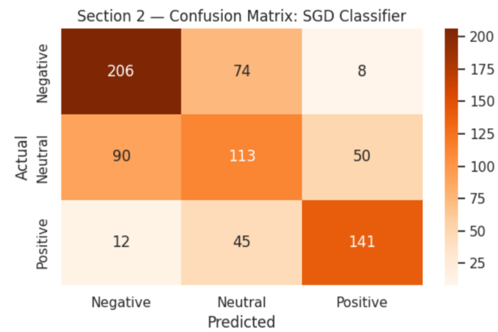

Fig. 35: SGD Classifier — British Airways Dataset

C. Dataset 3: Global Airline Reviews

Thirdly, there is a dataset conerning airline reviews around the world sourced from Kaggle database. This is because of the higher number of missing values across many of the ratings provided in different categories.

Therefore, the sentiment labels for the dataset were calculated based on only three dimensions: Seat Comfort, Cabin Staff Service, and Value For Money.

Table IV reports the performance of all five models on Dataset 3.

TABLE IV: Model Performance on Dataset 3 — Global Airline Reviews

Model	Accuracy	Precision	Recall	F1-Score
SGD Classifier	0.7534	0.7027	0.7534	0.7043
Logistic Regression	0.7527	0.7132	0.7527	0.7189
Naive Bayes	0.7397	0.6956	0.7397	0.6828
Random Forest	0.7308	0.5946	0.7308	0.6545
Decision Tree	0.7007	0.6567	0.7007	0.6678

The best results were obtained for dataset 3, where logistic regression model obtained an F1-Score value of 0.7189, and SGD Classifier yielded the best accuracy score of 0.7534.

The increased effeciency is explained by the fact that the language in the airline review was relatively cleaner due to the global nature of the collected reviews, along with more balanced sentiments.

These findings prove that the proposed approach to NLP works effectively for airlines regardless of review source. The confusion matrices for all five models on Dataset 3 are presented in Figures 36 through 40.

Logistic Regression

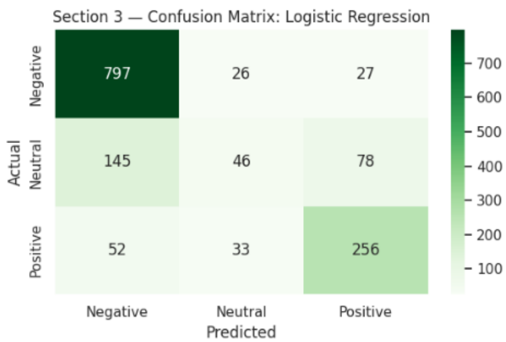

Fig. 36: Logistic Regression — Global Airlines Dataset

Decision Tree

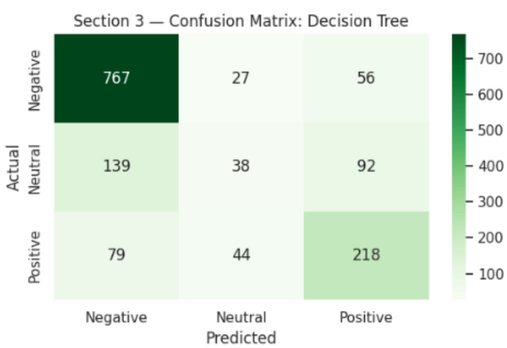

Fig. 37: Decision Tree — Global Airlines Dataset

Random Forest

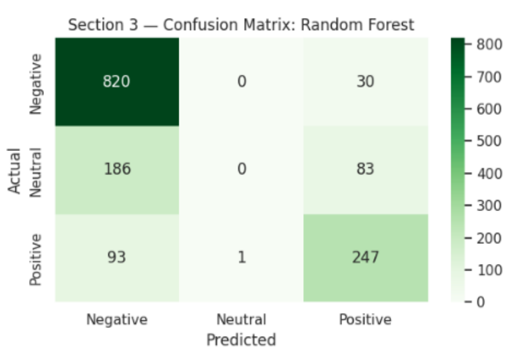

Fig. 38: Random Forest — Global Airlines Dataset

Naive Bayes

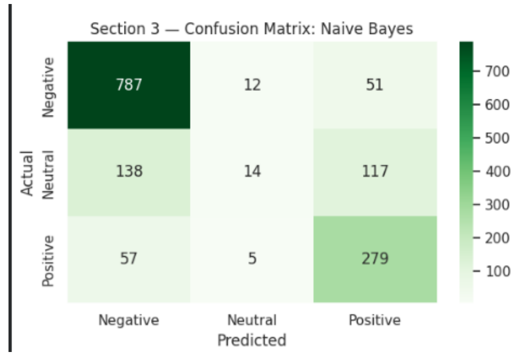

Fig. 39: Naive Bayes — Global Airlines Dataset

SGD Classifier

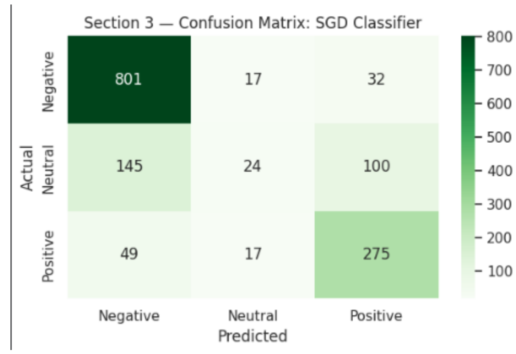

Fig. 40: SGD Classifier — Global Airlines Dataset

To evaluate the effectiveness of the showed sentiment models standard evaluation metrics we used: Accuracy, Precision, Recall, and weighted F1-Score These metrics provide a comprehensive assessment of classification performance by measuring prediction correctness, classification quality, sensitivity, and the balance between precision and recall. The weighted F1-Score was particularly important due to the class imbalance observed in several datasets

A grand comparative analysis was conducted to evaluate the performance of all machine learning algorithms across the airline review datasets used in this study. The experimental results obtained from each dataset were first combined into a unified dataframe to facilitate a consolidated comparison among all models. Each record was labeled according to its corresponding dataset, including the Skytrax, British Airways, and Global Airlines datasets.

Following the integration process, the combined dataframe was transformed into a long-format structure using dataframe melting techniques. This restructuring simplified the visualization and comparative analysis of multiple evaluation metrics across different models and datasets. The comparison focused primarily on Accuracy, Precision, Recall, and weighted F1-Score.

To enhance the interpretability of the experimental findings, separate bar charts were generated for each evaluation metric.

These visualizations provide a clear comparison of machine learning model performance across the three datasets and illustrate how dataset characteristics, sentiment distribution, and textual diversity influence classification outcomes. The graphical analysis also helps identify the most stable and consistently effective algorithms.

The comparative results demonstrated that Logistic Regression and SGD Classifier achieved the highest and most consistent performance across most datasets. In contrast, Naive Bayes and Random Forest produced moderate results that varied depending on dataset structure and feature distribution. Decision Tree consistently recorded lower performance compared to the other classification algorithms, indicating limited generalization capability for this sentiment analysis task.

Overall, the evaluation highlights the significance of selecting appropriate machine learning algorithms for sentiment analysis applications. The findings confirm that model performance is strongly influenced by dataset properties, class imbalance, linguistic diversity, and review structure. Therefore, comprehensive evaluation across multiple datasets and performance metrics is essential for developing reliable real-world sentiment classification systems.

The following figures present the comparative performance results for all machine learning models across the selected evaluation metrics.

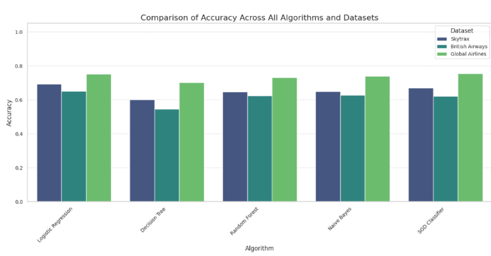

Fig. 41: Comparison of Accuracy Scores Across All Models

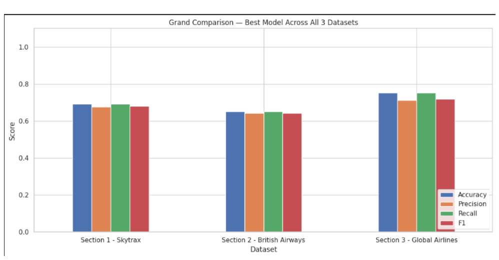

Fig. 42: Best Performing Model Across Datasets

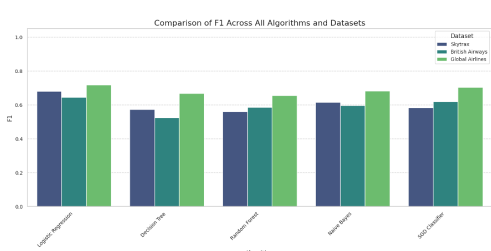

Fig. 43: Weighted F1-Scores Across All Models

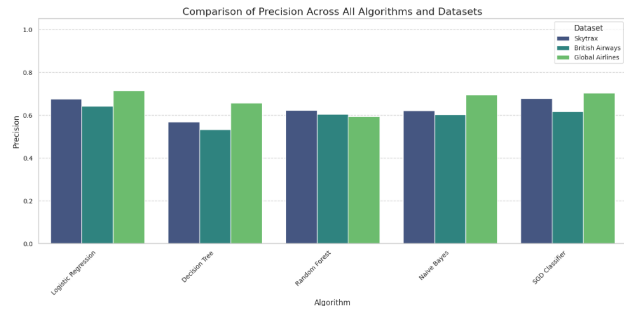

Fig. 44: Precision Scores Across All Models

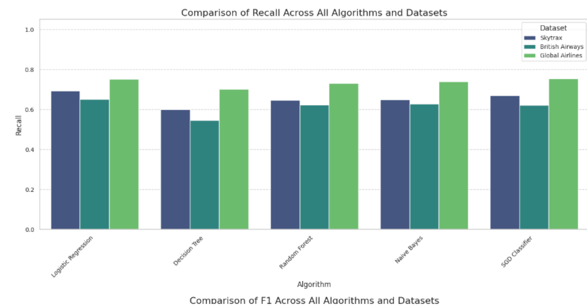

Fig. 45: Recall Scores Across All Models

V. Conclusion

This paper presented a structured NLP and machine learning framework for sentiment classification of airline passenger reviews across three real-world datasets, totaling over 50,000 reviews. Among the five evaluated models, Logistic Regression demonstrated the most consistent performance, achieving weighted F1-Scores of 0.6804 and 0.7189, while the SGD Classifier recorded the highest accuracy of 0.7534 on the Global Airlines dataset. Decision Tree ranked lowest due to overfitting, and the Neutral class remained the most challenging to classify across all experiments. The results confirm that a well-designed traditional machine learning pipeline can deliver reliable sentiment classification without the computational overhead of deep learning. Future work will explore transformer-based models such as BERT, SMOTE-based class balancing, and aspect-based sentiment analysis to further improve classification performance.

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